TERRITORY RUNNER — 10-Minute Team Presentation Master Sheet

`MAX TIME: 10:00 MIN`

Synchronized Speaking Schedule & Transition Handovers for College Evaluation Panel

0:00 – 3:00 (3 MINS) : AYUSH — Server-Authoritative Backend & Concurrency — SPEAKER 1

0:00 - 1:00 (Hook): Strava 1D passive tracks vs Territory Runner 2D organic GPS polygons. Core Rule: Mobile untrusted; Go 1.24 + PostGIS 3.4 is the single source of truth.
1:00 - 2:00 (Core Tech): Spatial concurrency: why OCC and Redis Redlock fail. Solution: `pg_advisory_xact_lock`. Kinematic anti-cheat speed ceilings (25 km/h run, 50 km/h bike, 4.5 m/s² accel).
2:00 - 3:00 (Integrity & Handover): Idempotency keys prevent double claims. 79 Go tests, sub-35ms p95 latency, Docker containerization.

> 🗣️ **Ayush's Handover Words:** "Now, I hand over to my teammate Gurudutt, who will present the mobile UI design and athletic design system."

3:00 – 6:00 (3 MINS) : GURUDUTT — Mobile UI/UX Design & Athletic Ergonomics — SPEAKER 2

3:00 - 4:00 (Design Philosophy): Outdoor runner constraints: direct noon sunlight glare & sweat/bouncing hands. Athletic Minimal design system: "Ink and Pacer Blue".
4:00 - 5:00 (Map HUD & Hold Gesture): Carbon `#090B0F` (anti-glare, OLED battery saver) + Pacer Blue `#4C8DFF` (conquest). 60 FPS vector map using Flutter `RepaintBoundary`. 1400ms Hold-to-Finish circular gesture with 350ms haptic pulses.
5:00 - 6:00 (Feedback & Handover): AppToast notification framework for reconciliation & GPS warnings. 317 Flutter tests, zero layout overflows, responsive across all screen densities.

> 🗣️ **Gurudutt's Handover Words:** "Now, I pass the floor to my teammate Avinash, who will present our computational geometry algorithms, loop detection, and project documentation."

6:00 – 9:00 (3 MINS) : AVINASH — Computational Geometry, Parity & Master Documentation — SPEAKER 3

6:00 - 7:00 (Lasso Closure): Organic loops cross track breadcrumbs. Lasso Closure Algorithm: 150m minimum guard, 25m snap buffer, line segment intersection math, Clipper2 polygon slicing.
7:00 - 8:00 (Parity & Donut Holes): Scaled planar Shoelace on Flutter vs PostGIS `ST_Area`: < 0.5% area discrepancy proven across 15 golden fixtures. Ray-Casting Donut Classifier (Jordan Curve Theorem) to exclude lakes.
8:00 - 9:00 (Documentation & Close): Master SRS, architectural sequence flows, API contracts, and cross-platform verification. Inviting examiner questions.

> 🗣️ **Avinash's Closing Words:** "In conclusion, Territory Runner unites server authority, athletic UI, and geometric rigor. My teammates Ayush, Gurudutt, and I are now ready for your questions."

9:00 – 10:00 (1 MIN) : INTEGRATED LIVE DEMO & TEAM TRANSITION TO VIVA — ALL THREE

• Show phone running the app in dark mode, trigger an instant loop closure on the map, show 317 Flutter + 79 Go tests passing badge.

Quick Viva Responsibility Routing (Who Answers What):

Topic Asked by Examiner	Answering Member	Core Keywords to Say in First 5 Seconds
Concurrency, DB, Deadlocks, Redis	**AYUSH**	"pg_advisory_xact_lock inside PostgreSQL kernel, auto-released on commit/rollback."
Anti-Cheat, Vehicle Spoofing, GPS Jumps	**AYUSH**	"Kinematic sliding window, 25 km/h run ceiling, 4.5 m/s² acceleration ceiling."
UI Glare, OLED Battery, Color Tokens	**GURUDUTT**	"Ink #090B0F carbon canvas, Pacer Blue #4C8DFF, high-contrast sunlight readability."

Topic Asked by Examiner	Answering Member	Core Keywords to Say in First 5 Seconds
Accidental Button Touches, Map 60 FPS	**GURUDUTT**	"1400ms hold-to-finish with haptic ticks; Flutter RepaintBoundary layer isolation."
Loop Detection, Self-Intersection, Clipper2	**AVINASH**	"Lasso proximity snapping (25m buffer), segment line intersection math, tail pruning."
Shoelace Parity, Geodesics, Donut Holes	**AVINASH**	"Scaled planar Shoelace matching PostGIS ST_Area within 0.5%; Jordan Curve ray-casting."